Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/07/2023
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Round Robin - Thermophysical Property Measurement		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	04/01/2017	**End Date:**	02/29/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 02/29/2024 per PI (Ed., 4/4/22).		
Key Personnel Changes/Previous PI:	NOTE: Per the PI, CoInvestigator Mikhail Krivilev from Udmurt State University is no longer with the project. (Ed., 2/2/23)		
COI Name (Institution):			
Grant/Contract No.:	NNX17AH41G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Researchers go to space to conduct material property measurements that are required to model terrestrial manufacturing processes to provide higher performance and discover new materials to make people's lives better around the world. We justify the expense of doing tests in microgravity by concentrating on those materials which are impossible to evaluate on Earth or have properties that can only be measured with higher precision when gravitational accelerations are minimized. We know we get better results but we do not yet know how much better. The JAXA (Japanese Aerospace Exploration Agency) Electrostatic Levitation Furnace (ELF) was designed to process high temperature oxides but it can be readily adapted to allow testing of industrially important metals and alloys as well. By selecting materials that are already being processed using another technique, the European Space Agency (ESA) Electromagnetic Levitation facility on the International Space Station (ISS), these two facilities can be compared so that researchers know their relative strengths such that they may complement each other to maximize science output from space tests. Our objectives are to conduct space experiments to compare to ground results on a baseline material set and to model the influence of gravity on measurement techniques by various research groups across the world. Our approach is to employ Round Robin testing by multiple researchers on the same material using different techniques, on different facilities, with different analysis protocols to compare results. The significance of this work is that we will then know the relative increase in measurement fidelity and which material works best in which facility to efficiently use the limited available space sciences resources such as launch cost, crew time, and telemetry bandwidth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This program leverages the existing ESA International Topical Team organization to develop a strategy to compare the performance of current flight and ground facilities investigating thermophysical property measurement using levitation techniques. New experiments are proposed for the JAXA-ELF facility on the ISS to complement other ongoing investigations that use both the ISS-based Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) and ground-based facilities. We will efficiently use available resources, both fiscal and crew time, by selecting materials compatible with those previously prioritized by other existing research groups. In doing so, we will expand the capabilities of space electrostatic levitation processing to allow for investigation of additional thermophysical material properties and manifesting new material classes across microgravity platforms. The key scientific questions we address are: (1) How do we control and minimize gravity-induced systematic error in thermophysical property measurement and does theory explain the observed deviations in results? (2) For which material classes are space-based platforms delivering thermophysical property measurement results with higher accuracy and/or precision than ground-based facilities? Thermophysical property testing in microgravity provides enhanced fidelity of results purportedly due to three main effects. First, without strong gravitational accelerations and with reduced levitation forces a more spherical sample allows for better analysis of experimental behavior. Second, better control of convection in space results in higher measurement precision. Third, sedimentation and buoyancy induced segregation are eliminated in reduced gravity. But how much better do we do in space as compared to ground? The obvious solution is to quantify variability across platforms – both in space and on ground.
Task Progress:	Highlights of Work Accomplished During Grant Period Mar 2022- Feb 2023. • Graduated 2 MS and 1 BS students with 3 thesis submissions. • Submitted 3 peer-reviewed articles & 3 presentations (see Bibliography). • Conducted flight testing of the final 3 flight samples during International Space Station (ISS) expedition 64 using the Japanese Space Agency (JAXA) electrostatic levitation furnace (ELF) facility. • Honored as Fellow for the American Society for Materials 2022. • Featured speaking engagements at: Japanese Society for Microgravity Applications-34, Asian Microgravity Symposium, Kyoto University of Advanced Science, Korea University, Leibniz Institute for Solid State and Materials Research in Dresden (IFW-Dresden), and University of Michigan. • Completed and submitted the final science report for this contract to the NASA Task Book.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Abstracts for Journals and Proceedings	Nawer J, Ishikawa T, Oda H, Koyama C, Pichler P, Leitner T, Pottlacher G, Matson DM. "A comparison of Inconel 625 density and thermal expansion coefficient measurements in terrestrial and microgravity environments." 13th Asian Thermophysical Property Conference (ATPC), Sendai, Japan, September 26-30, 2022. Abstracts. 13th Asian Thermophysical Property Conference (ATPC), Sendai, Japan, September 26-30, 2022. , Sep-2022
Abstracts for Journals and Proceedings	Nawer J, Ishikawa T, Oda H, Koyama C, Matson DM. "Quantifying facility performance during surface tension and viscosity measurement of liquid Zr." 13th International Workshop on Subsecond Thermophysics, Karlsruhe, Germany, June 8-10, 2022. Abstracts. 13th International Workshop on Subsecond Thermophysics, Karlsruhe, Germany, June 8-10, 2022. , Jun-2022
Articles in Peer-reviewed Journals	Mohr M, Dong Y, Bracker GP, Hyers RW, Matson DM, Zboray R, Frison R, Dommann A, Neels A, Xiao X, Brillo J, Busch R, Novakovic R, Srirangam P, Fecht HJ. "Electromagnetic levitation containerless processing of metallic materials in microgravity: Thermophysical properties." npj Microgravity. 2023 May 2;9:34. https://doi.org/10.1038/s41526-023-00281-4 , May-2023
Articles in Peer-reviewed Journals	Nawer J, Ishikawa T, Oda H, Saruwatari H, Koyama C, Xiao X, Schneider S, Kolbe M, Matson DM. "Uncertainty analysis and performance evaluation: a measurement of thermophysical properties of liquid Au in microgravity." npj Microgravity. 2023 May 24;9:38. https://doi.org/10.1038/s41526-023-00277-0 , May-2023
Dissertations and Theses	Cai Z. "Droplet deformation and oscillation in an acoustic levitator." MS Thesis, Tufts University, May 2022. , May-2022
Dissertations and Theses	Fakhrul M. "Applying a wave-like nucleation model to determine inoculant potency in powder feedstock." MS Thesis, Tufts University, February 2023. , Feb-2023

Dissertations and Theses	Waskow D. "The effect of undercooling in an Fe-Ni alloy." BS Thesis, Tufts University, February 2023. , Feb-2023